

22.3 A 42Gb/s Single-Ended Hybrid-DFE PAM-3 Receiver for GDDR7 Memory Interfaces

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Demand for high-bandwidth graphics memory is rapidly increasing for various applications such as AI, deep learning, autonomous cars, etc. To meet this demand, GDDR6X uses multilevel signaling (PAM4) for the first time in DRAM [1], while GDDR7 targets data-rates 28Gb/s or higher using PAM3 signaling. However, the increased data rate incurs higher inter-symbol interference (ISI), while PAM3 is more vulnerable to ISI than PAM2 signaling. Decision feedback equalizers (DFEs) are widely used in receivers (RX) to remove post-cursor ISI without amplifying noise. A PAM3 RX, with an analog DFE, typically employs a direct feedback (DF) implementation, which must meet stringent feedback timing requirements. As the target speed increases it is challenging to reduce the feedback time to within 1 unit interval (UI) for a PAM3 RX due to the increased line parasitics and gate loading. While the loop-unrolled (LU) DFE technique mitigates feedback timing constraints, implementing a PAM3 LU DFE results in additional area and power overhead. In view of these drawbacks, this paper proposes a new type of PAM3 DFE, which combines the qualities of DF and LU DFEs to minimize area and power overhead, while also reducing the feedback time.

Figure 22.3.1 (a) shows a typical block diagram for priorDOF PAM3 DFEs [2], [3]. The left figure shows a DFE that only compensates for outer signal-level transitions (corresponding to worst-case) to have a shorter feedback time. However, ISI caused by the small transitions (corresponding to non-worst-case) remains, which degrades the voltage margin especially when ISI is limiting the BER. On the other hand, the DFE in Fig. 22.3.1(a) right is compensating for all transitions; thus, it is effective even for large ISI, but it suffers from a longer feedback time due to increased parasitics and gate loading on the feedback path. As shown in Fig. 22.3.1(b), employing LU DFE in PAM3 RX requires six reference voltages, and the number of sense



amplifiers (S/A) and voltage-to current converters (V2I) increases threefold. Additionally, a 3 : 1 MUX is required, leading to a significant increase in both area and power consumption.

[Figure 22.3.2](#) shows the proposed hybrid PAM3 DFE. The key idea of the proposed DFE is that ISI due to large and small transitions are removed in different manners. More specifically, among the six possible transitions that change levels in PAM3, small transitions (4 cases) are compensated using DF DFE, while large transitions (2 cases) are compensated using LU DFE. The DF DFE operation utilizes the outputs of CTLE_H and CTLE_L separately, with each encountering only a single-gate load. This configuration allows for the feedback path to be formed efficiently, thereby minimizing the feedback time. For the LU DFE operation, the DFE strength is fixed to $V_{REF-H-3}$ and $V_{REF-L+3}$ for LU High /Low, respectively, to compensate for ISI. In both DF and LU DFE operations, the final output is determined by using the output of the opposite-side S/A as the SEL signal for the multiplexer. By connecting the V2I to 1, the loading on the CTLE output is reduced. To compensate for large transitions, the LU operation has 3× the DFE strength compared to the DF DFE operation, which is accomplished by adjusting the size of the V2I. The proposed design reduces the number of blocks to 2/3 compared to a conventional LU DFE, requiring only a 2:1 rather than a 3:1 multiplexer. Overall, this minimizes the area and power overhead while reducing the feedback time.

The detailed operation of the hybrid PAM3 DFE is shown in [Fig. 22.3.3](#). For a 00 to 11 transition where S/A DF is high, since the MSB of the previous data $D[n-1]$ is 0, input data is sampled at $V2I-1$. In S/A LU High, regardless of the $D[n-1]$ level, the input data is sampled at $V2I-3$. The type of transition, major vs minor, is determined by the LSB output: if it is 0 then a large transition occurred, and if it is 1 then a small transition occurred. Therefore, the LSB output is used as the SEL signal to control MUX_H. Therefore, in this case, since $SEL_H = 0$, the output of S/A LU High is selected. In the case of transition from 01 to 11, since $D[n-1]$ MSB is also 0, in S/A DF High samples at $V2I-1$. However, in this case, since $D[n-1]$ LSB is 1, it signifies a small transition, thus the data sampled at S/A DF High is selected. In the final case where there is a transition from 11 to 01, since $D[n-1]$ MSB is 1, the data is sampled at $V2I+1$ in S/A DF High. Since $SEL_H = 1$, the MUX selects the data sampled at DF S/A $V2I+1$ over the data from LU S/A $V2I-3$.

The proposed hybrid PAM3 DFE reduces the feedback time but has larger area and power consumption compared to a DF DFE due to having more blocks. Therefore, the criteria for determining the optimal DFE type based on the target speed is presented in [Fig. 22.3.4](#). When the target speed is low, both hybrid DFE and DF DFE satisfy the feedback-time requirements, and the DF DFE type, which consumes less power, is advantageous. However, as the target speed increases the power consumption of the DF DFE starts increasing rapidly: the size of the S/A and local clock drivers must be increased to meet the reducing feedback time. On the other hand, loading of the feedback path in the hybrid DFE is lower than that in DF, allowing the feedback time to be met with a smaller increase in power consumption. Therefore, both hybrid and DF DFE increase power consumption to meet the feedback time. However, in cases where hybrid DFE consumes less power than DF DFE, using hybrid DFE is more advantageous. In the final case, if the target speed increases further, there can be a situation where, regardless of increasing the size, DF DFE cannot satisfy the feedback time. In this case, there is no choice but to use hybrid DFE, which may be able to meet the feedback time.

[Figure 22.3.5\(a\)](#) compares the simulation results of the feedback time between DF and hybrid PAM3 DFE with respect to the S/A input swing. The hybrid PAM3 DFE can reduce the metal line of feedback path length by 63% compared to a conventional PAM3 DF DFE, and the DFE input gate loading is halved. For the hybrid PAM3 DFE, the feedback time meets the 476ps threshold required for 42Gb/s operation, when the input swing is 40mV or higher. However, for DF DFE, the feedback time fails to meet this requirement, regardless of how much the input swing increases. [Figure 22.3.5\(b\)](#) presents the result of a 42Gb/s write Shmoo simulation for an off-chip environment with an insertion loss of -5dB at Nyquist. Since the feedback time of the hybrid PAM3 DFE satisfies the operation frequency, the write eye size improves with each DFE step.

The measurement Shmoo plot and BER for an RX with the proposed hybrid PAM3 DFE at 42Gb/s are shown in [Fig. 22.3.6\(a\)](#) and [\(b\)](#). All measurements use a PRBS11 pattern. For a BER of 10^{-9} , the hybrid PAM3 DFE achieves a valid window of 858mV/14ps, 79.2mV/14ps(0.29UI) with an optimal DFE setting. For a BER of 10^{-12} , the horizontal eye-opening is 0.27UI. A comparison table of prior graphics DRAM receivers [4], [5] is shown in [Fig. 22.3.6\(c\)](#). A negative-capacitance equalizer (NCE) is used to enhance the

bandwidth of the CTLE. Compared to a GDDR6 receiver, the area of the previous GDDR7 receiver and this work increases by 15% and 48%, but power efficiency decreases by 19% and 1 %. In this work, the feedback time is reduced by 20%, enabling 42Gb/s operation. [Figure 22.3.7](#) presents the micro-photograph of the prototype chip fabricated in a 1 b-nm DRAM process.



Figure 22.3.1:
(a) Conventional direct-feedback PAM3 DFE and (b) a loop-unrolled PAM3DFE.



Figure 22.3.2:
Hybrid PAM3 DFE's (a) block diagram, (b) sense amplifier and voltage-to-current converter.



Figure 22.3.3:
Hybrid PAM3 DFE operation. Data is sampled simultaneously with different DFE strengths at two S/As, DF and LU. Depending on the data transition, one of the two S/A outputs is selected.



Figure 22.3.4:
Selection criteria for the PAM3 DFE, depending on the target speed.



Figure 22.3.5:
Simulation results: (a) feedback time post simulation. (b) 42Gb/s write shmoo.



Figure 22.3.6:
42Gb/s measurement results using a PRBS11 pattern: (a) Shmoo plot at 10-9 BER and (b) horizontal bathtub curve. (c) Comparison to prior work.



Figure 22.3.7:
GDDR7 hybrid-PAM3 DFE receiver chip micrograph.

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